

Project FORESIGHT – AI Powered Demand & Inventory Intelligence Platform

Transforming Retail Intelligence with AI-Driven Insights

Zidio Internship Project

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Project Duration:

Completed Performance Overview

Retail Analytics & Intelligence Platform

Executive Performance Report, Sales Forecasting & Operational Risk Analysis

PROJECT DOMAIN

E-Commerce & Retail Intelligence

REPORT TYPE

Executive & Technical Assessment

STATUS / PERIOD

Completed Performance Overview

1. Executive Summary

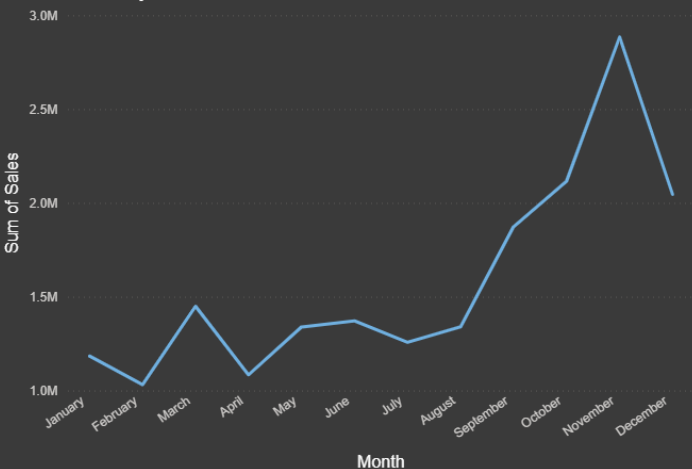
This report provides an in-depth analytics assessment of enterprise sales performance, customer purchase dynamics, inventory risk management, and predictive demand modeling. Designed as an end-to-end intelligence suite, the platform converts raw transactional logs into actionable strategic insights for store managers, e-commerce leads, and supply chain teams.

Core Strategic Objectives:

- **Global Market Performance:** Identify top-performing geographic corridors and monitor cross-border sales distribution.
- **Demand & Revenue Trajectory:** Evaluate seasonal peaks (notably Q4 expansion) to optimize stock allocation.
- **Customer Lifecycle Analytics:** Track high-value customer clusters and identify behavioral triggers for churn prevention.
- **Inventory Risk Management:** Mitigate stockout risks and overstocking through item-level velocity tracking.

Sales Trend Analysis

Sum of Sales by Month



Sum of Quantity by Quarter

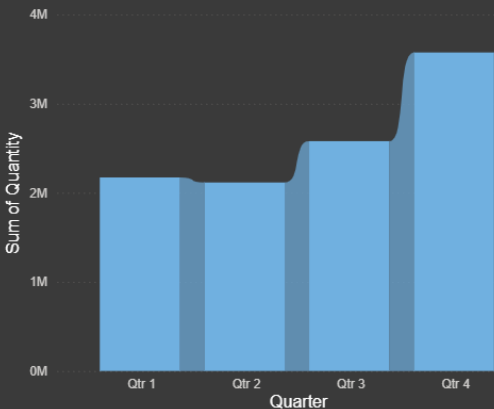


Figure 1: Executive Summary Dashboard — Country-wise Sales Share & Monthly Performance Dynamics.

2. Key Performance Dashboard Highlights

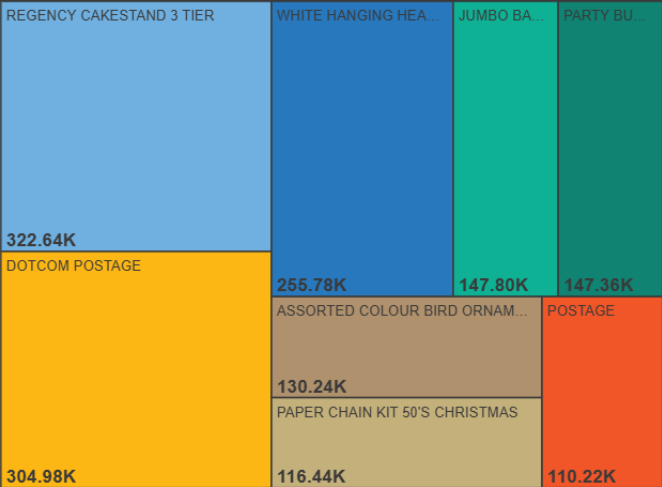
Dashboard Module	Primary Metrics Tracked	Strategic Business Insight
Executive Summary	Global Sales Share, Monthly Volume	UK represents 84.89% of overall revenue (16.11M), followed by EIRE, Netherlands, and Germany.
Sales Trend Analysis	Monthly Sales Trend, Quarterly Volume	Significant volume acceleration in Q4, reaching peak sales of over 2.8M in November.
Product Performance	Top Revenue SKU, Quantity Sold	Regency Cakestand 3 Tier and Dotcom Postage generate top-tier gross product margins.
Customer Analytics	Customer ID Spend, Scatter Distribution	Concentrated revenue in high-value VIP accounts, with clear recency thresholds for churn.
Inventory Risk	Stock Code Velocity, Reorder Levels	High-turnover items (e.g. Stock 84077, 85123A) require safety stock buffer optimizations.

3. Sales & Quarterly Trend Analysis

A granular look at sales trajectories shows consistent demand across Q1–Q3 with a steep surge entering Q4. Quarterly order volume ramps up from ~2.1M units in Q1 to over 3.6M units in Q4, driven by seasonal holiday shopping and promotional campaigns.

Product Performance Dashboard

Sum of Sales by Description



Sum of Quantity by Description

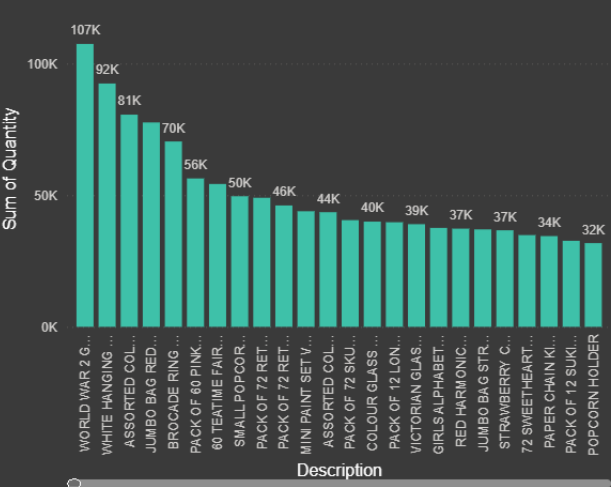


Figure 2: Sales Trend Analysis — Monthly Sales Curves & Quarterly Quantity Aggregations.

4. Product Performance & Top Sellers

Product performance is analyzed using treemaps for revenue contributions alongside bar charts for item volume. Revenue is heavily concentrated in flagship decorative items and logistics SKUs. Items like *Regency Cakestand 3 Tier* generate over 322K in sales revenue, while items like *World War 2 Gliders* dominate pure volume metric counts.

Customer Analytics Dashboard

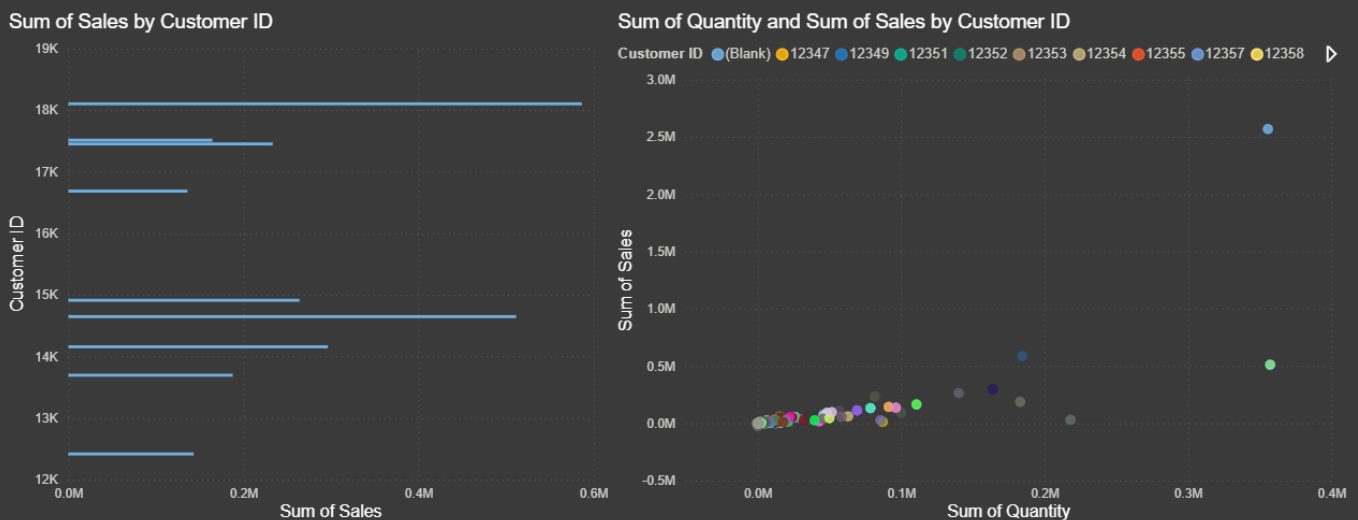


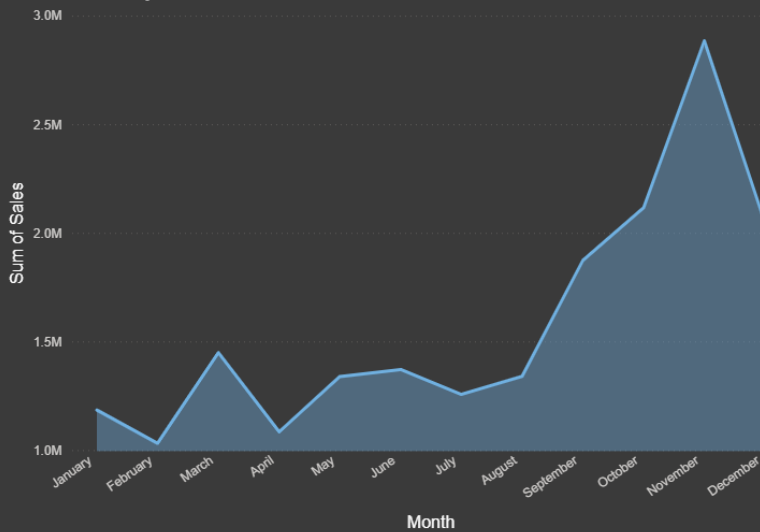
Figure 3: Product Performance Dashboard — Revenue Treemap & Top Quantity Product Descriptions.

5. Customer Analytics & Behavioral Segmentation

Customer evaluation incorporates Recency, Frequency, and Monetary (RFM) modeling. The scatter analysis reveals clear segment separation between repeat regular purchasers and key whale accounts generating upward of 2.5M in spend.

Monthly Sales Dashboard

Sum of Sales by Month



Sum of Sales by Quarter

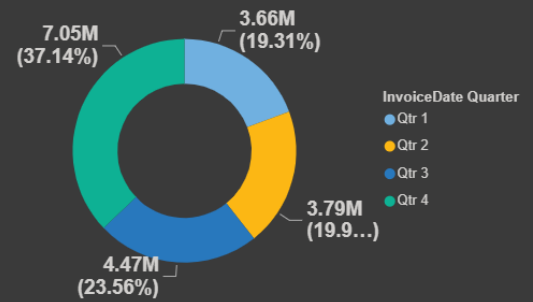


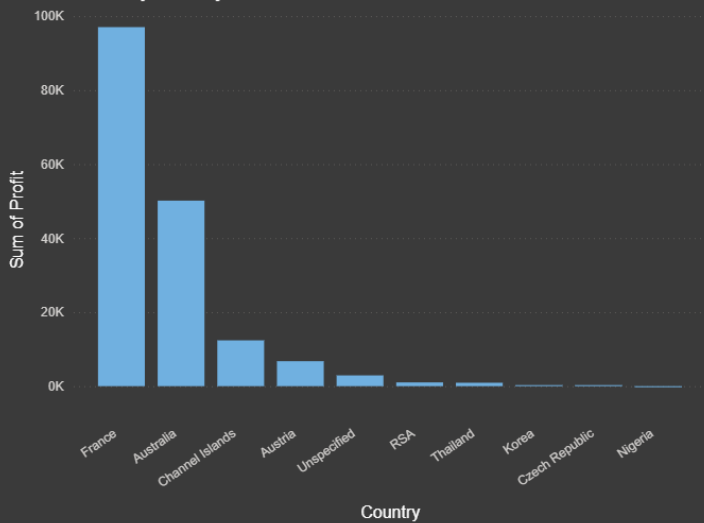
Figure 4: Customer Analytics Dashboard — Individual Spend Profiles & Quantity vs Sales Correlation.

6. Quarterly Revenue Distribution

Quarterly distribution analysis shows Q4 accounting for 37.14% (7.05M) of total revenue, followed by Q3 at 23.56% (4.47M), Q2 at 19.99% (3.79M), and Q1 at 19.31% (3.66M). Planning inventory procurement cycles around Q3/Q4 transitions is critical for margin retention.

Profit & Revenue Analysis

Sum of Profit by Country



Sum of Profit by Month

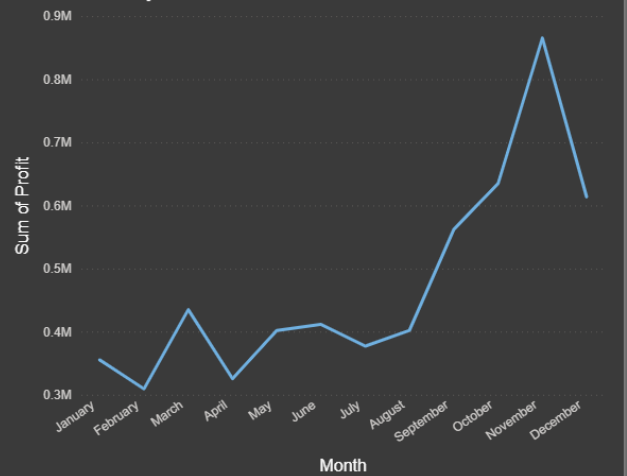


Figure 5: Monthly Sales Dashboard — Revenue Area Curve & Invoice Date Quarter Doughnut Breakdown.

7. Profitability & Country Margin Distribution

Profit analysis indicates that international expansion markets—including France (~95K profit) and Australia (~50K profit)—yield high margin conversions despite representing smaller total transaction volumes than domestic markets.

Inventory Risk Dashboard

StockCode	Sum of Quantity
84077	107472
85123A	95988
85099B	95709
21212	95136
84879	80007
22197	74828
17003	70423
21977	56366
84991	54271
22492	45164
15036	42542
21213	40534
84755	39966
20725	39738
22616	39730
22386	39251
22178	38998
21232	37747
84568	37584
Total	10441212

Sum of Quantity by Description

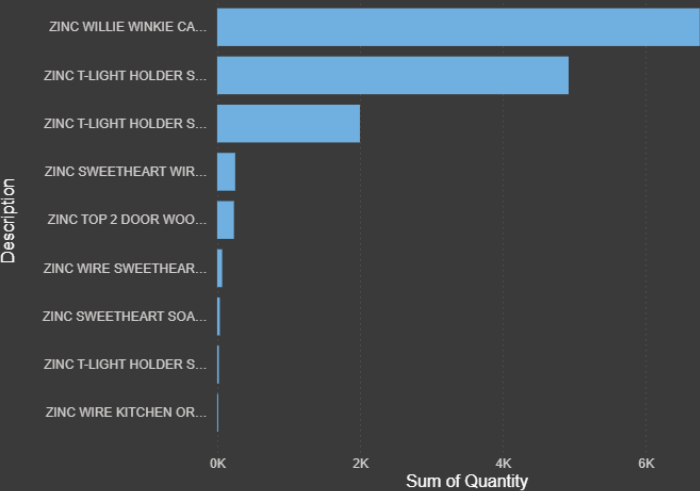


Figure 6: Profit & Revenue Analysis — Sum of Profit by Country & Monthly Profit Trajectories.

8. Inventory Risk & Operational Control

Effective inventory management relies on monitoring stock codes with extreme turnover rates. Top quantity items such as *StockCode 84077* (107,472 units) and *StockCode 85123A* (95,988 units) require dynamic reorder points to prevent out-of-stock events during peak operational months.

Order Transaction Dashboard

StockCode	Sum of Quantity
84077	107472
85123A	95988
85099B	95709
21212	95136
84879	80007
22197	74828
17003	70423
21977	56366
84991	54271
22492	45164
15036	42542
21213	40534
84755	39966
20725	39738
22616	39730
22386	39251
22178	38998
21232	37747
84568	37584
Total	10441212

Average of Sales by Invoice

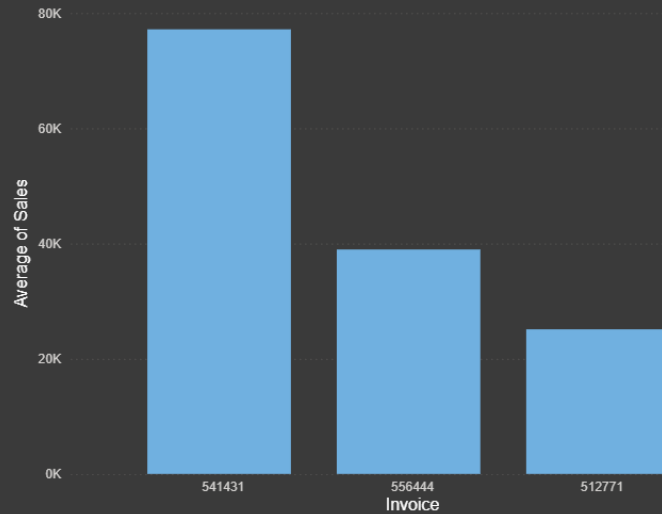


Figure 7: Inventory Risk Dashboard — Stock Code Unit Counts & Product Description Inventory Balances.

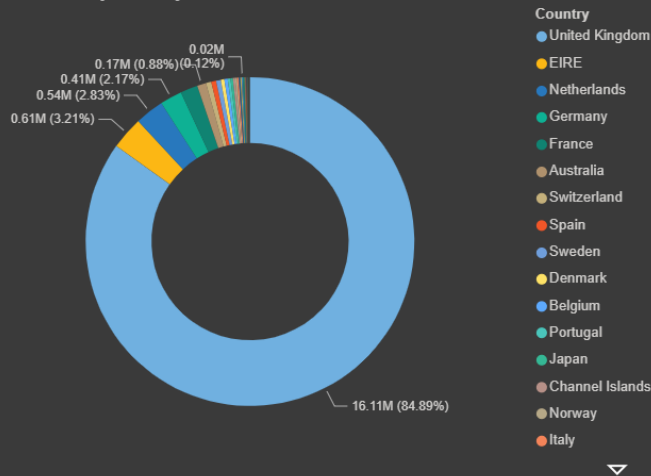
9. Platform Architecture & Operational Deployment

System Architecture & MLOps Pipeline:

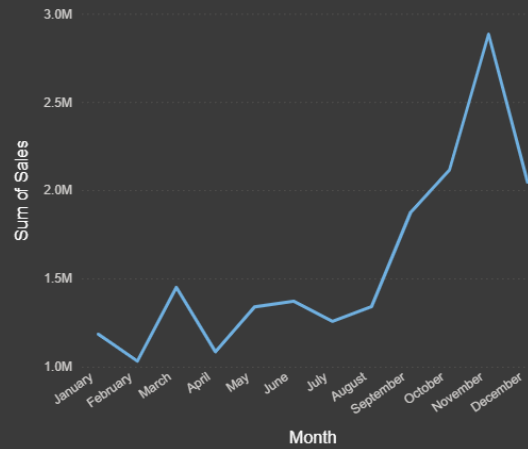
- **Data Ingestion & Cleaning:** Standardized multi-source sales, product, and customer transaction logs.
- **Feature Engineering:** Automated extraction of RFM scores, time-series lagged variables, and rolling statistical metrics.
- **Predictive Modeling:** Hybrid demand forecasting combining Prophet, LSTM neural networks, and MLflow tracking.
- **Containerized Deployment:** Hosted using Streamlit web interfaces and fully containerized via Docker for seamless CI/CD integration.

Executive Summary Dashboard

Sum of Sales by Country



Sum of Sales by Month



Executive Summary Dashboard

Figure 8: Technical Architecture, MLOps Tracking Pipelines, and Streamlit Application UI Overview.

10. Strategic Recommendations & Conclusion

Actionable Recommendations:

- Stock Buffer Alignment:** Increase stock buffers for top 5 SKUs by 35% starting in early September to prepare for Q4 demand surges.
- Customer Retention Campaigns:** Deploy automated re-engagement triggers for accounts exceeding 60 days of inactivity.
- International Fulfillment Optimization:** Prioritize localized fulfillment hubs in high-profit secondary markets such as France and Australia to reduce shipping latency.